



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Charge and Current — $Q = It$

One equation, one trap. The trap is the units.

Key concept

CURRENT IS A FLOW OF CHARGE

Current is not “electricity”. It is the **rate at which charge flows** — how much charge goes past a point each second.

$$\text{charge flow} = \text{current} \times \text{time} \qquad Q = It$$

Q = charge flow, in **coulombs (C)** · I = current, in **amps (A)** · t = time, in **seconds (s)**

Rearranged: $I = \frac{Q}{t}$ and $t = \frac{Q}{I}$

The trap, and it is the one that costs marks. t must be in **seconds**. Real questions give you *minutes*, or hours. Convert **first**, and write the converted number down so a marker can see it.

1 minute = 60 s · **2 minutes** = 120 s · **1 hour** = 3600 s

Worked example

A current of 0.25 A flows for **3 minutes**. Calculate the charge that flows.

Convert first. $t = 3 \times 60 = 180 \text{ s}$ $Q = It = 0.25 \times 180 = \mathbf{45 \text{ C}}$

Using $t = 3$ would give 0.75 C — sixty times too small, and no method mark for a conversion you never wrote down.

1. State what is meant by *electric current*. [1]

2. Write down the equation that links charge flow, current and time. [1]

3. A doorbell draws a current of 0.85 A for 40 seconds. Calculate the charge flow. **Give your answer to 2 significant figures.** [3]

4. A current of 1.75 A flows in a heater element for 80 seconds. Calculate the charge flow. [2]

5. There is a current of 0.045 A in an LED for 240 seconds. Calculate the charge flow. [2]

6. A charge of 624 C flows through a motor in 130 s. Calculate the current. [2]

7. A phone charger supplies a current of 0.90 A for **2 minutes**. Calculate the charge that flows. [3]

8. A charge of 1500 C flows through a lamp in **5 minutes**. Calculate the current. [3]

9. A torch bulb carries a current of 0.24 A. How long must it be switched on for 72 C of charge to flow? Give your answer **in minutes**. [3]

10. A student says: “A *bigger current always means more charge has flowed.*” Explain why this is not necessarily true. [2]

11. A single closed loop carries a current of 2.0 A. Explain why the same charge flows past every point in that loop in one minute. [2]

Extension — how big is a coulomb, really?

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Background (not on the AQA specification — you will not be asked for this)

You have spent this sheet calculating charges of 34 C, 108 C, 140 C. Reasonable-sounding numbers. But **how much charge is one coulomb?**

A single electron carries a charge of

$$e = 1.6 \times 10^{-19} \text{ C}$$

which is a fantastically small amount. So the number of electrons needed to make up a charge Q is

$$\text{number of electrons} = \frac{Q}{e}$$

It is here because a coulomb is otherwise just a word — and because the answer is genuinely surprising.

Worked example

The doorbell in question 3 moved 34 C of charge. How many electrons is that?

$$\text{number} = \frac{Q}{e} = \frac{34}{1.6 \times 10^{-19}} = \mathbf{2.1 \times 10^{20}} \text{ electrons}$$

That is 210,000,000,000,000,000 electrons — for one press of a doorbell.

12. A torch bulb carries a current of 0.30 A for 60 seconds. Calculate the charge that flows, and then the number of electrons. [3]

13. A current of 5.0 A flows in a kettle element for 2 minutes. Calculate the number of electrons that pass through it. [3]

One more surprise, and it is worth knowing. Those electrons are not racing along the wire. In a copper wire carrying a normal current they *drift* at well under a millimetre per second — slower than you can walk, slower than a snail.

So why does a lamp light the *instant* you close the switch?

14. Explain why a lamp lights immediately when the switch is closed, even though the electrons themselves move extremely slowly. [3]

A straw already full of peas: push one in, one drops out of the far end at once — and no single pea travelled the straw.